Admin

Your system nearing completion -- exciting!

Interrupts

Today

Exceptional control flow

Suspend, jump to different code, then resume

How to do this safely and correctly

Low-level mechanisms

Next lecture

Using interrupts as client

Coordination of activity

Concurrency, preemption, multiple handlers, shared data

Blocking I/O

```
while (1) {
  char ch = keyboard_read_next();
  update_screen();
}
```

How long does it take to send a scan code?

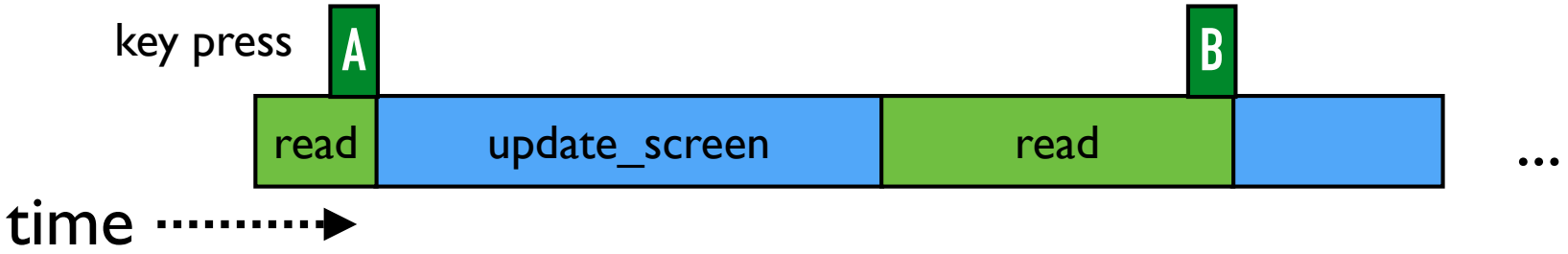
11 bits, clock rate 15kHz

How long does it take to update the screen?

What could go wrong?

Blocking I/O

```
while (1) {
  char ch = keyboard_read_next();
  update_screen();
}
```



Blocking I/O

```
while (1) {
  char ch = keyboard_read_next();
  update_screen();
}
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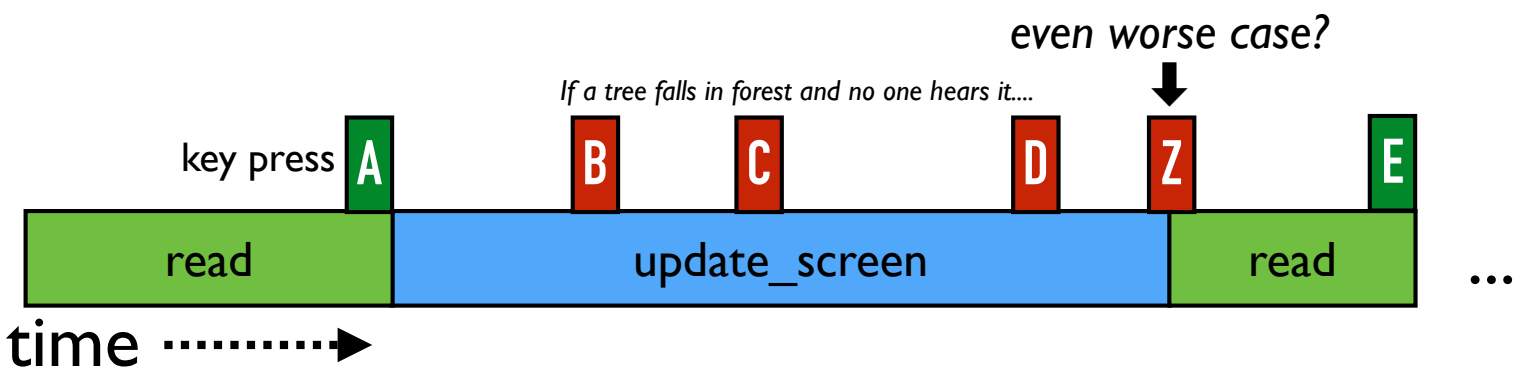


The Problem

Ongoing, long-running tasks (graphics, simulations, applications) keep CPU occupied, but …

When an external event arises, need to respond quickly.

Consider: Why does your phone have a ringer? What would you have to do to receive a call if it didn't?

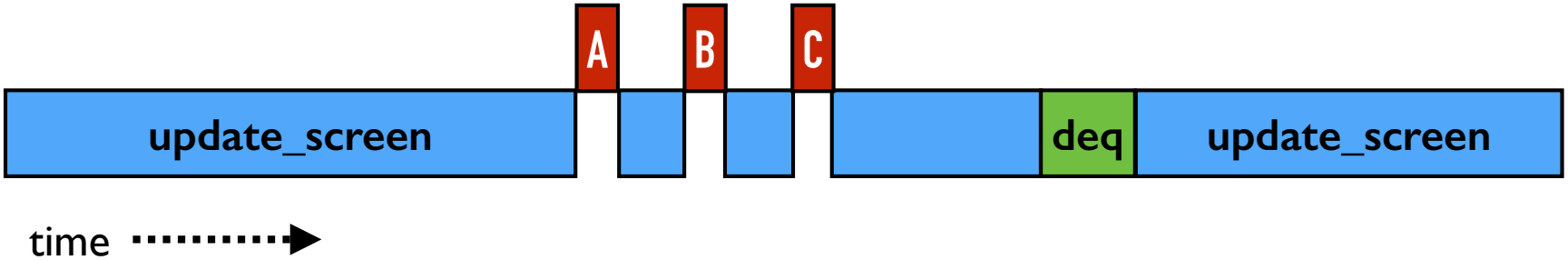
Concurrency

How to juggle doing more than one thing at once?

```
while (1) {
   update_screen();
   dequeue scancode from queue
}
```

```
when a scancode arrives {
  add scancode to queue;
}
```

scancodes buffered for later processing



Interrupts to the rescue!

Processor pauses current execution, switch to handle interrupt, then resume execution

- Asynchronous external event (peripherals, timer)
- Synchronous exception (invalid address, illegal instruction)

Critical for responsive systems, hosted OS

Interrupts are essential and powerful, but getting them right requires using everything you've learned:

- Architecture, assembly, linking, memory, C, peripherals, …

code/button-blocking
code/button-interrupt

Interrupted control flow

static volatile int gCount;

```
void update_screen(void)
{
  console_clear();
  for (int i = 0; i < N; i++)
    console_printf("%d", gCoun
}
```

```
void button_pressed(uintptr_t pc, void *p)
{
    gCount++;
    gpio_interrupt_clear(BUTTON);
}
```

9 9 9 9 9 9
9 10 10 10
10 10

Suspend current activity, execute other code, then resume, ... this will be tricky!

Interrupt mechanics

Somewhat analogous to function call

- Suspend currently executing code, save state
- Jump to handler code, process interrupt
- When finished, restore state and resume

Must adhere to conventions to avoid stepping on each other

- Consider: processor state, register use, memory
- Hardware support helps out

C abstraction ...
Asm understanding!

```
update_screen:
  add sp,sp,-16
  sd  ra,8(sp)
  sd  s0,0(sp)
  add s0,sp,16
  jal 0x400006c4 <console_clear>
  lui a0,0x40008
  add a0,a0,496
  jal 0x40000224 <console_printf>
  li  s1,0
  j   0x400002cc <update_screen+68>
  lui a5,0x40022
  lw  a1,-760(a5)
  lui a0,0x40008
  add a0,a0,528 # 0x4000
  jal 0x40000224 <conso
```

Interrupt!

How to go from **update_screen** to **button_pressed** and back again?

- pause
- preserve state
- execute other code
- resume

```
button_pressed:
 add sp,sp,-16
 sd  ra,8(sp)
 sd  s0,0(sp)
 add s0,sp,16
 lui a4,0x40022
 lw  a5,-760(a4)
                gpio_interrupt_clear>
 ret
```

The RISC-V Instruction Set Manual
Volume II: Privileged Architecture
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Terminology

Exception

- Unusual condition when executing instruction (invalid memory address, illegal instruction)

Trap

- Synchronous transfer of control to trap handler

Interrupt

- External event that occurs asynchronously
- Executing instruction is interrupted, will experience a trap

Processor modes

Machine (most privileged), Supervisor, User (least)

Reset starts in **machine** (M) mode
(all our code executes in M mode)

Trap usually executes with more privilege
(hardware support to change mode on enter/exit trap handler)

What code is executed on trap?

- `mtvec` CSR stores address of code to jump to

CSRs

MXLEN-1	MXLEN-2 36	35 34	33 32	31 23	22	21	20	19	18	17
SD	**WPRI**	SXL[1:0]	UXL[1:0]	**WPRI**	TSR	TW	TVM	MXR	SUM	MPRV
1	MXLEN-37	2	2	9	1	1	1	1	1	1

16 15	14 13	12 11	10 9	8	7	6	5	4	3	2	1	0
XS[1:0]	FS[1:0]	MPP[1:0]	**WPRI**	SPP	MPIE	**WPRI**	SPIE	UPIE	MIE	**WPRI**	SIE	UIE
2	2	2	2	1	1	1	1	1	1	1	1	1

Figure 3.7: Machine-mode status register (`mstatus`) for RV64.

MXLEN-1 12	11	10	9	8	7	6	5	4	3	2	1	0
WPRI	MEIE	**WPRI**	SEIE	UEIE	MTIE	**WPRI**	STIE	UTIE	MSIE	**WPRI**	SSIE	USIE
MXLEN-12	1	1	1	1	1	1	1	1	1	1	1	1

Figure 3.12: Machine interrupt-enable register (`mie`).

```
interrupts_global_enable:
    li a0,1<<3
    csrs mstatus,a0
    li a0,1<<11
    csrs mie,a0
    ret
```

Interrupt, hardware-side

External event triggers interrupt. Processor response:

- Complete current instruction

Pause, transfer control

- Save interrupted **pc** into **mepc** CSR, elevate privilege
- Disable further interrupts
- Execution jumps to **mtvec** CSR

Software takes over (trap handler)

process/clear interrupt

trap handler must save/restore any registers it uses

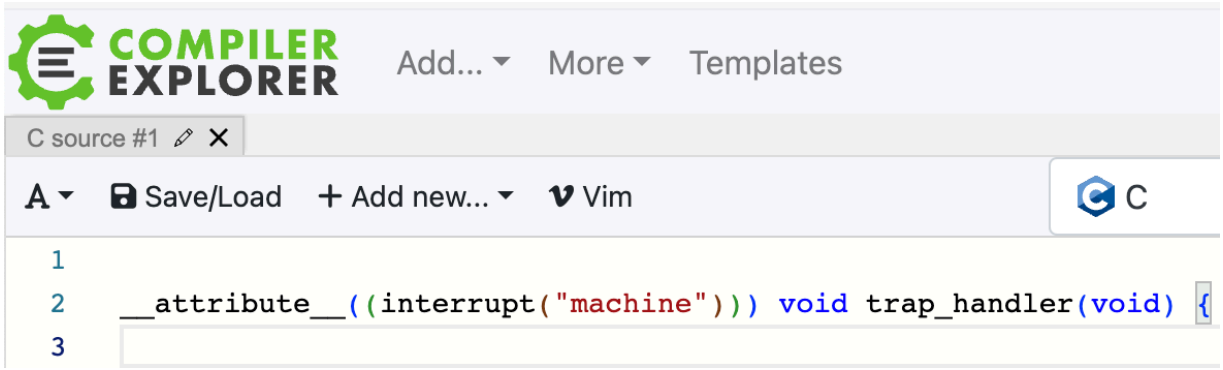
Resume interrupted execution

mret instruction will pc = mepc, restore privilege level, reenable interrupts

Interrupt, software-side

```
__attribute__((interrupt("machine")))
void trap_handler(void)
{
   ....
}
```

How does this attribute change the generated code? Let's find out!



Install trap handler

```
__attribute__((interrupt("machine"))) void trap_handler(void) {
  // code to respond to interrupt
}
```

```
void main(void) {
    set_mtvec(trap_handler); // store function pointer
}
```

```
set_mtvec:     // asm.s
    csrw mtvec,a0
    ret
```

Interrupts (so far)

Hardware support

- Processor modes, exceptional control flow

Software init

- Install handler, enable source

Exception received, handler must:

- Init stack, preserve registers, handle event
- Restore and resume

Next Lecture

Which interrupts are supported and how to configure
(events from gpio, timer, uart, ...)

How to manage dispatch to handler per event source

What steps needed to init/enable interrupts

Writing safe interrupt handlers
How to share state without step on each other